



MA 765: Fractional Differential Equations

Course Code	MA 765
Course Name	Fractional Differential Equations
L-T-P-C	4-0-0-4
Prerequisites	Real and Functional Analysis
Intended for	Ph.D.
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Basic Theory of fractional differential equations**

Definition of fractional derivative, Riemann Liouville, Caputo derivatives, Existence and uniqueness of solutions, dependence of solutions on initial conditions, General order fractional differential equations.

Text and Reference Books

1. Diethlem, K., **The Analysis of Fractional Differential Equations**, Springer, 2010.
2. Podlubny, I., **Fractional Differential Equations**, Academic press, 1999.
3. Dumitru Baleanu, José António Tenreiro Machado, Albert C. J. Luo, **Fractional Dynamics and Control**, Springer, 2012.
4. Vasily E. Tarasov, **Fractional Dynamics**, Springer, 2010.